

A brief history

Before getting into the different steps involved, let's take a look at the brief history of fabrication.

Back when feature widths were way bigger than 10 micrometer, purity wasn't as big an issue as it is today in semiconductor manufacturing. However, as more and more circuits got integrated, the cleanrooms – the spaces of manufacture – became cleaner than before. These days, fabs get pressurized using filtered air so that even the tiniest particles could be removed. These particles, if left unchecked may come to rest on the wafers, leading to defects. Also, the workers in a fabrication facility these days have to wear special suits so that the devices could be protected from human contamination.

During the early days, in the 1960s, device manufacturing was largely an American enterprise – with companies in Texas and California on the forefront. But now, the global business that it is, it has spread to other parts of the world including Asia and the Middle East. It's common enough for the world's leading semiconductor manufacturers to have facilities around the world. For instance, Intel- the biggest manufacturer in the world owns facilities not just in the US but also in Asia and Europe.



Cleanrooms everywhere

Some of the top manufacturers in present-day world include Samsung(Korea), IBM(US), SMIC(China) and Texas Instruments(US) to name a few.

The wafers

A typical semiconductor wafer is made from pure (extremely pure) silicon. This silicon is turned into mono-crystalline cylindrical ingots. The diameter of these ingots could be up to 300 mm. They are sliced to form the wafers. Each wafer would be about 0.75 mm thick.

The steps involved

Processing

In fabrication, the different processing steps come under four catego-